CRT reciprocity and the polynomial block radical

November 13, 2026. Unsigned producer evidence [pcc:unverified].

General component. Over any field let F, H be monic, $a = \deg F$, $b = \deg H > 0$, $b + 1 < a$, and suppose supplied polynomials obey

$$UF + V(fH) = 1, \qquad \mu(Z) = [X^{a-1}]((ZV) \bmod F).$$

For $\deg A \le a$, $\deg B \le b$, Lean characterizes the full block pairing:

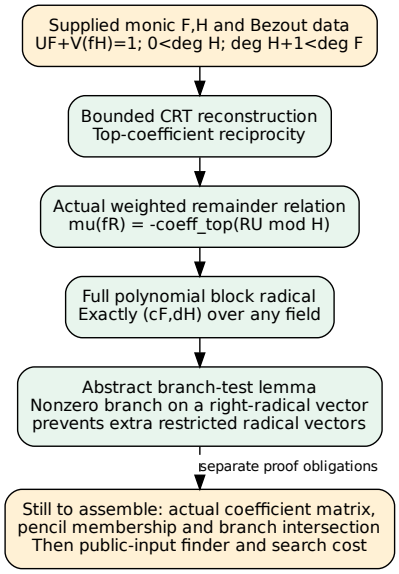
$$\mu(AQ - fBT) = 0 \quad \text{for all } \deg Q \le a,\ \deg T \le b \quad \Longleftrightarrow \quad (A, B) = (cF, dH)$$

for scalars c, d. This uses the actual quotient pairing, not a rank assumption. No characteristic or separability condition is imposed. The even norm chart $a = b + 2$ fits; constant H is excluded here.

Core proof. Bounded CRT remainders satisfy $SH + TF = P$: the difference is divisible by FH but has smaller degree. Comparing the top coefficient gives reciprocity. The weighted instance is

$$\mu(fR) + [X^{b-1}]((RU) \bmod H) = 0, \qquad \deg R < a + b - 1.$$

The existing coefficient detector shows that a unit-weighted quotient pairing annihilates P exactly when its modulus divides P. At the bounded degree, this is exactly scalar span. Applying this separately to the upper and lower blocks gives the displayed radical. For full lower tests, $\deg(BT) \le 2b < a + b - 1$ explains the strict gap.



Branch restriction, evidence and remaining assembly

Branch-test component. For a bilinear pairing $\mathcal{B}$ and linear functional ℓ, suppose k is in the right radical and $\ell(k) \neq 0$. Then vanishing against all tests in $\ker \ell$ is equivalent to vanishing against all tests. Indeed every v can be replaced by $v - \ell(v)\ell(k)^{-1}k$. This proves no new left-radical vectors arise under that explicit condition. Identifying the actual polynomial branch, its scalar intersection and coefficient matrix remains an assembly task.

Proof map. `NormCRTReciprocity` supplies reconstruction, coefficient comparison, canonical remainders and weighted reciprocity. `NormPairingRadical` supplies ideal annihilators, bounded/full upper and lower radicals, `normBlockRadical` and `branchTestsDetectFull`. Exact hypotheses, fourteen unsigned claim bindings, sources and raw receipts are linked from the accompanying Markdown report.

Checks. Lean4.35.0 with unchanged pinned CryptoLean/Mathlib. Frozen source commits `3487f067d` (CRT) and `0d1208d8b` (radical). The first gate covers95 declarations before the radical module was tracked; the final gate covers104 with zero failing and zero trusted declarations. Both import-inclusive fresh replays exited0; their separate statuses are retained. Only standard foundations `propext`, `Classical.choice` and `Quot.sound` are admitted. No signed PCC verdict is claimed. The first radical build failed on an overbroad rewrite; subsequent builds correct it. Failed and completed logs are preserved separately.

Native record `IDC1h4b52a3733d292783` checks the universal residual

$$(PV - FQ)H + (PU - HT)F - P = P(UF + VH - 1) - FH(Q + T).$$

The unchanged issuer/checker uses 32 statement-derived and 32 extra-stream points, with exact universal proof-source hash. It does not execute Lean or establish conditional degree or radical conclusions. Adapter tests:22 pass, including inherited tamper/statement-swapping controls. No timing claim.

Relation to rank-directed completion. The earlier proposal sought $W_{\text{new}} = W_{\text{base}} + C(T)H(T)$ with reduced crossing rank. Quotient moments give a concrete alternative form with a characterized radical. Reachability through that completion family and an efficient choice of its free coefficients are not established by these results.

Next obligations. Identify actual bounded coefficient matrices, normalization factors, $\mu(G) = 0$ and coefficient-pencil membership; instantiate the branch lemma and corank1 recovery; cover degree drops. Then construct moments from public data with accounted $2^{o(n)}$ bit cost, $n = \lceil \log_2 r \rceil$. Current Bézout data is supplied with the witness.

Sources and resources. Exact pinned Mathlib monic-division, coefficient, degree and coprime declarations; previous detector and next-proof note. KB unavailable; bounded review, no priority claim for classical CRT. No paid job, empirical timing/scaling campaign, benchmark or ecdb update. Monthly isolab budget remains exhausted. Editable DOT/SVG/PDF diagram and source-linked Markdown report accompany this PDF.